

Welcome

Welcome to ENAE601

Course meets Tuesday, Thursday 9:30-10:45 AM in [ATL 1106](#)  (<https://25live.collegenet.com/pro/umc#!home/location/2649/details>) and online simultaneously through Zoom; recordings will be available sometime after each day's lecture ends through Panopto. Note: regardless of section of registration, all students can participate online or in class (subject only to seating capacity in classroom).



Course content

Astrodynamics, see [syllabus](#)  (<https://go.umd.edu/ENAE601-2026-syllabus>)

Astrodynamics is the study of the motion of spacecraft; for most of the class, it will be the motion dominated by a single gravitating body; at the end, we will cover the n-body problem. Observations and orbit determination are important for spacecraft navigation, but not included in this course (see ENAE606/788U spring 2027). Some numerical computations can be performed by a Python package that wraps [Orekit](#)  (<https://www.orekit.org/>), a library for astrodynamics. This will require the installation of Python and a few very common packages (AstroPy and NumPy chiefly).

Textbook: Orbital Mechanics for Engineering Students, Revised Reprint 4th Edition - August 31, 2020, Imprint: Butterworth-Heinemann, Author: Howard D. Curtis, Paperback ISBN: 9780128240250, eBook ISBN: 9780323853453

Other references to be introduced when the topic is covered.

Course requirements

- Homeworks (about 7) - 30%
- Midterm exam, take-home mid-late October - 35%
- Project, information forthcoming in October, due at end of semester - 35%

Homeworks and exam will be turned in through Gradescope.



Your Instructor



 Dr. L. Healy

 Teacher

 healy@umd.edu

 **Before class, in the classroom**